

Discussions on Health, Diet, Exercise & Nutrition

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What is Balanced Nutrition

Balanced nutrition is like a well-managed investment portfolio - **Variety, Proportion, and Balance.**

If you invest everything in one stock—say, carbohydrates—you might crash.

We need to diversify to survive



American Dietary Guidelines 2025 - 2030



ICMR – Food Plate



Macro Nutrients



A large orange circle on the left side of the slide.

Calorie Breakdown per Gram

- **Carbohydrates:** 4 Kcalories
- **Protein:** 4 Kcalories
- **Fat:** 9 Kcalories

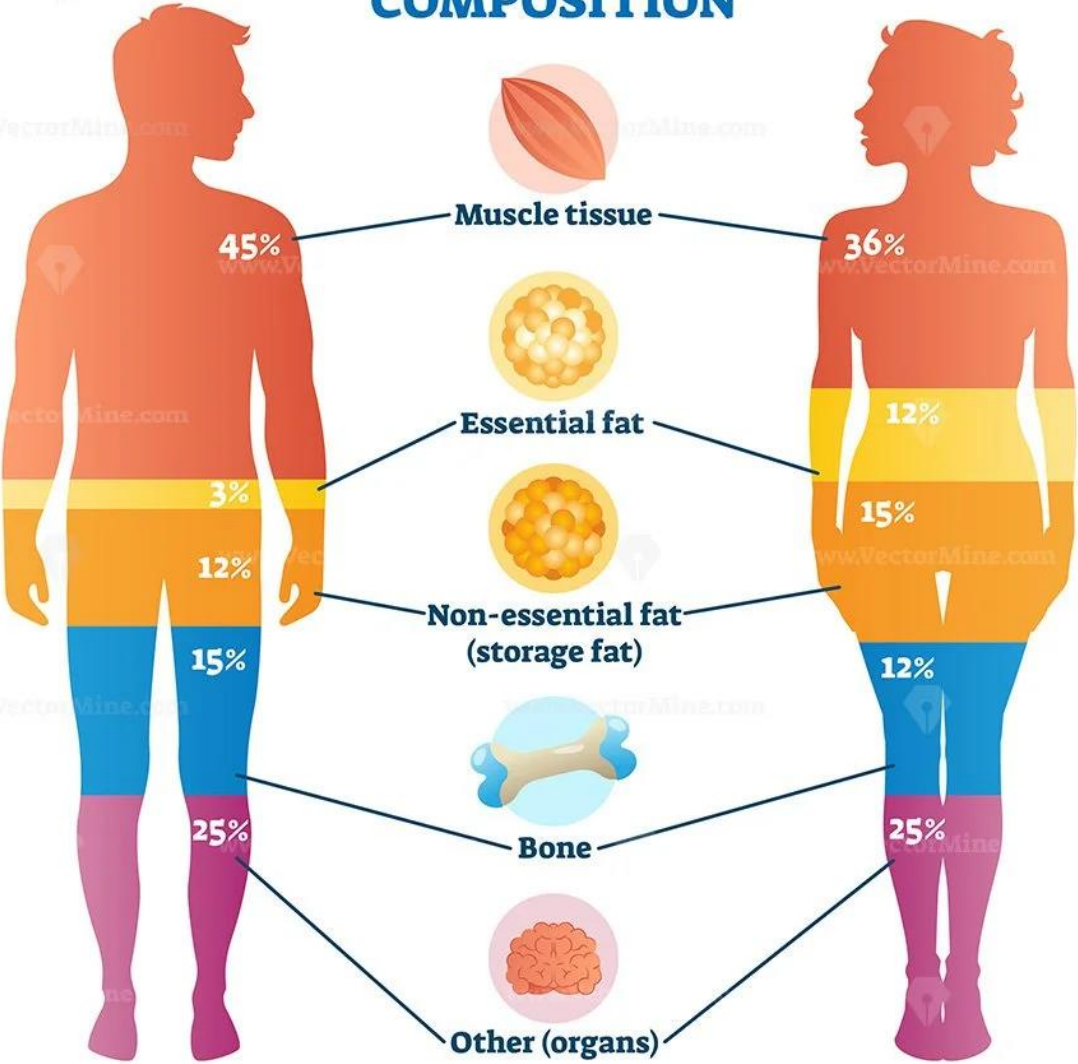


Sr No	Food Item	Recommended Quantity (g/ml)	Calories (Kcal)	Substitute
1	Atta or Rice or Atta and Rice	620	2100	Semolina/cornflakes/wheatflakes/ Daliya 450g, Bread 8-10 slices
2	Dal	90	400	Rajma/Kabuli chana/Dried peas/ Black chana 90g
3	Fresh Vegetable	200	60-140	Tinned vegetable 150g/ Dehydrated Vegetable 100g
4	Potato	210	140	Sweet potato 400g/ Potato 400g
5	Onion	60	100	
6	Fresh fruit	250	80-100	Dried fruit 200g/ Fruit juice 250ml
7	Milk	250	155	Curd 250ml/ Buttermilk 300ml
8	Meat/Chicken/Egg/ Fish	180	120-350	Eggs 3-4 Nos/ Paneer 180g/Tofu 200g/ Soya chunks 80g
9	Oil	60	540	Butter 50g
10	Sugar	60	240	Honey 60ml/ Jam 50g
11	Dry fruit	25	120	
	Total Calories:		4065-4495 Kcal	

Dietary Reference Intake per day
Ration Scale in the Indian Navy

Practical Aspects of Nutrition

HUMAN BODY COMPOSITION



	Male	Female	
Primary Mass	Higher Muscle (LBM)	Higher Essential Fat	Males need more total calories.
Protein	Higher Absolute Need	Lower Absolute Need	Males need more total grams to maintain muscle.
Fuel Source	Higher Carb Oxidation	Higher Fat Oxidation	Females often thrive on slightly higher fat ratios.
Iron	8 mg/day	18 mg/day	Females need more red meat, beans, or spinach.

GOOD CARBS



BAD CARBS



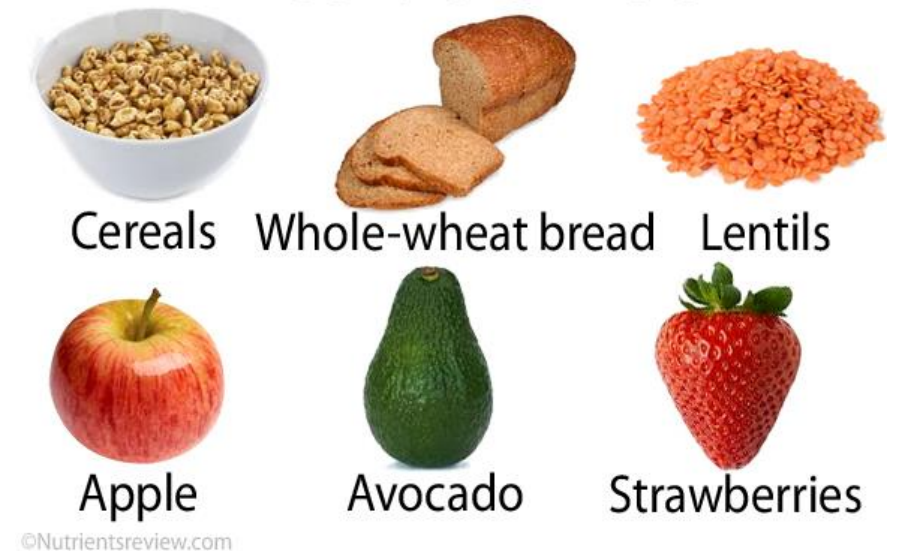
Fiber

- Soluble fiber - slows digestion and makes feel full after eating and soften stools
- Insoluble fibre - adds bulk to stool, preventing constipation.
- Low Glycemic Index and Low glycemic load – Good for Diabetics
- Lowers LDL – protects heart
- Support the immune system - protect against inflammation.
- Reduce the risk of developing dementia

Soluble Fiber



Insoluble Fiber



Macronutrients and Dietary Fibre in various food groups (per 100 gms raw weight)

Foods	Protein (g)	Fat (g)	Carbo hydrates (g)	Energy (Kcal)	Total dietary fibre (g)
Cereals	9.3	1.2	72	343	6
Millet	9.9	2.7	65	330	7
Pulses	22.8	3.0	49	323	12
GLVs	3.8	0.7	5	45	2
Roots & tubers	1.5	0.2	12	59	2
Vegetables	1.8	0.4	5	35	2
Nuts	17.5	41.3	18	516	9
Fruits	1.0	0.6	11	59	2
Meat & poultry	20.8	6.8	0	250	0
Fish & sea foods	18.4	3.1	2	110	0
Milk	3.1	4.2	5	72	0
Egg	13.3	10.0	1	147	0
Dry spices	8.5	10.0	31	240	17
Milk products	21.6	18.6	16	337	0
Dry fish	55.5	5.0	1	271	0
Cooking oil/fats		100.0		900	
Table sugar	0	0	100	400	0

Source – Indian Food Composition
Table 2017 - NIN

Glycemic Index (GI)

Measures how quickly food can increase blood glucose levels.

Does NOT measure portion size .



Glycemic Load (GL)

Measures how quickly food can increase blood glucose levels.

Considers GI & portion size.

Food Values: Glycemic Index/Glycemic Load

	Low GI	Med GI	High GI
Low GL	All-bran cereal (8,42) Apples (6,38) Carrots (3,47) Peanuts (1,14) Strawberries (1,40) Sweet Corn (9,54)	Beets (5,64) Cantaloupe (4,65) Pineapple (7,59) Sucrose, i.e. table sugar (7,68)	Popcorn (8,72) Watermelon (4,72) Whole wheat flour bread (9,71)
Med GL	Apple juice (11,40) Bananas (12,52) Fettucine (18,40) Orange juice (12,50) Sourdough wheat bread (15,54)	Life Cereal (16,66) New potatoes (12,57) Wild rice (18,57)	Cheerios (15,74) Shredded wheat (15,75)
High GL	Linguine (23,52) Macaroni (23,47) Spaghetti (20,42)	Couscous (23,65) White rice (23,64)	Baked Russet potatoes (26,85) Cornflakes (21,81)

Source: Revised International Table of Glycemic Index (GI) and Glycemic Load (GL), *The American Journal of Clinical Nutrition*, July 2002

GL in Fruits vs Fruit juice

Fruit Type	Whole Fruit (GI)	Fruit Juice (GI)	Difference
Orange	43	50–53	Juice is slightly higher; and also lacks pulp fiber.
Apple	36	41–44	Whole apple has a much lower GL due to skin fiber.
Pineapple	59	66+	Both are medium-to-high; juice spikes faster.
Grapefruit	25	45	Whole grapefruit is one of the lowest GI fruits.
Watermelon	76 (High GI)	80+	GL is low (8) for whole fruit because it's mostly water.



WHITE BREAD



PASTA



BREAKFAST CEREAL



BISCUITS



CAKES



WHITE RICE



PASTRIES



CRACKERS



Bad or Unhealthy Carbs – High GI and High GL

Fats – Good, Bad and Ugly



Poly Unsaturated Fatty Acids (PUFA)/ Mono Unsaturated Fatty Acids (MUFA) - Good

- Fish, Sea Foods, Eggs, Leafy Vegetable, Seed Oils, Cereals, Millets
- Cleans up the artery
- Helps to increase HDL



Saturated Fatty Acids - Bad

- Increase LDL, Total Cholesterol
- Increase inflammation
- Reduce insulin sensitivity

- Coconut oil, Palm oil, Ghee, Butter, Lard, Milk

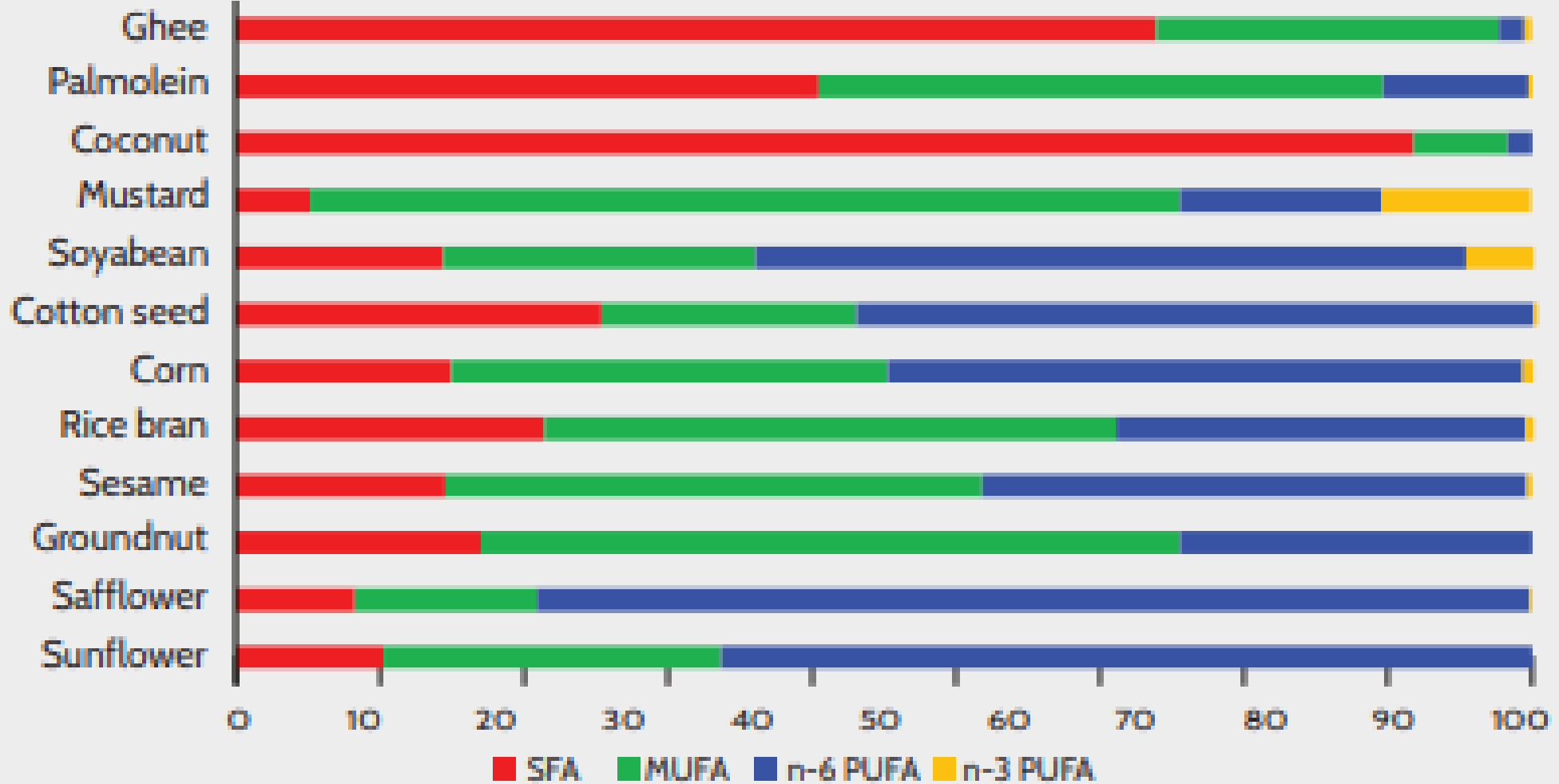


Transfats

- Transfat - Ugly
- Re - Heating of vegetable oils, bakery items, Deep Fries, Processed Foods
 - Raises bad cholesterol and lowers the good – hypertension/heart disease
 - decreases insulin sensitivity – diabetes
 - Increased risk of cancer



Fatty Acid Composition of Vegetable Oils



Source - NIN

How to Use Cooking Oils

- **Rotation:** olive oil, canola oil, groundnut oil, mustard oil, soybean oil, sunflower oil, rice bran oil - so you get the benefits of each.
- **High in MUFAs (Good for Sautéing/General Cooking):** Olive Oil (78%), Almond Oil (73%), Canola Oil (58%).
- **Good for Deep Frying (Higher Smoke Points):** Sesame, Rice Bran, or Groundnut Oil are recommended for deep frying.
- **Best for Salads/Dressings (Low Smoke Point/Flavorful):** Extra Virgin Olive Oil or Flaxseed Oil.



Protein

- Quality of protein (Biological Value) - depends on its ability to supply essential amino acids.
- Egg protein and animal protein high Quality (Biological Value)
- Plant food lower quality (Biological Value)
- Cow Milk Protein better than Soya/Almond Milk Protein
- Indian diets optimum protein
 - Balanced mixture of animal and vegetable proteins
 - Mix of cereals and pulses has all amino acids
- 12-15% protein is optimum (40-60gms per day)

Emerging Functional Foods - supporting health beyond basic nutrition

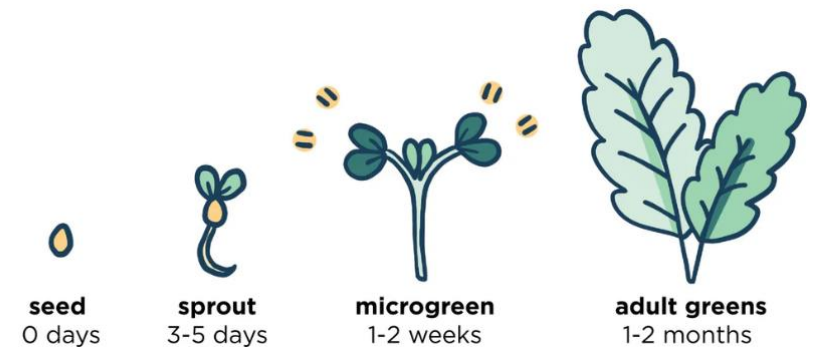
Prebiotics & Probiotics

Prebiotics (Food) act as nutrition for gut bacteria (Probiotics, helping them to flourish.



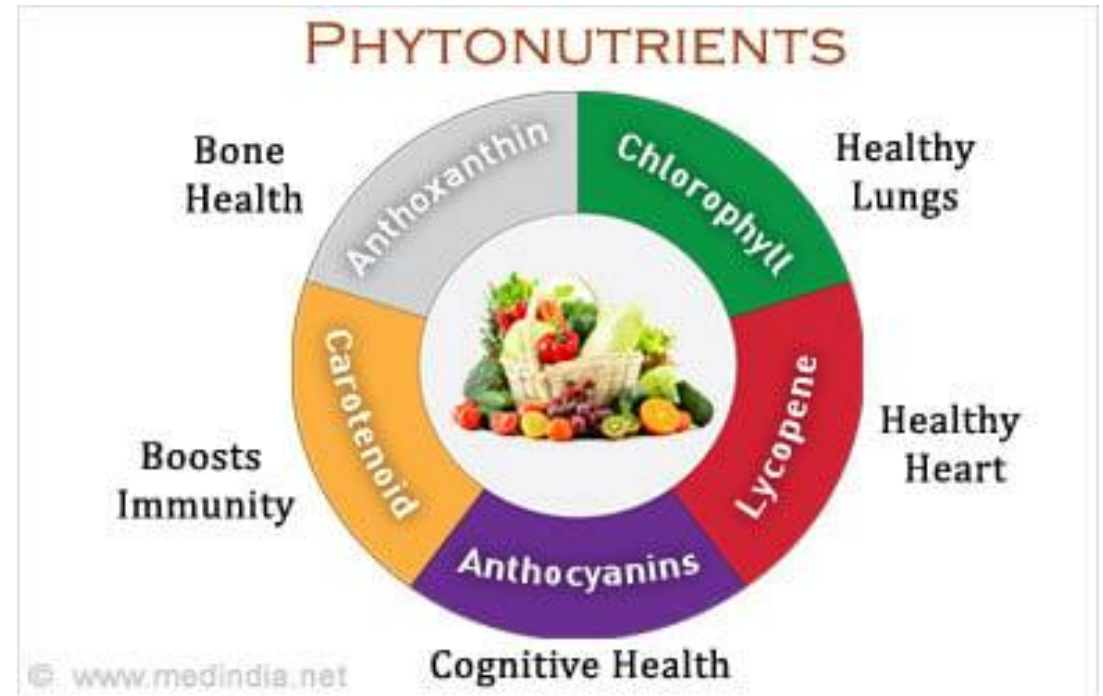
What are Microgreens? – Small Plant Big Nutrition

- Young, edible seedlings of vegetables and herbs, harvested just after the first true leaves develop, within a week or 10 days after sowing the seeds
- Offer concentrated flavors, vibrant colors,
- Dense nutrients like vitamins, antioxidants, and minerals
- Popular as garnishes or in salads
- Example – Carrots, Cilantro, Basil, Cabbage, Mustard
- Different from Sprouts and Baby Greens



What are Phytonutrients? – Eat the Rainbow

- Carotenoids (Anti Cancer, Anti Diabetes) – Carrots, Mango, Tomato
- Flavonoids (Antioxidant, Anticancer) - berries, dark chocolate, citrus fruits, green tea
- Saponins (Anti Diabetic, Liver Protective) – beans and legumes
- Phytic Acid (Anti Cancer) – Legumes
- Lignans (Hair regeneration, Mimic Oestrogen, Prevent Osteoporosis) – Pumpkin Seeds, Flax Seeds





Mood & Sleep "Bio-Hacks"

Tyrosin – Mood Enhancer

HOW TO GET MORE *TYROSINE*

- 
- SPIRULINA
 - SPINACH
 - EGGS (EGG WHITES)
 - SALMON
 - COTTAGE CHEESE
 - FAVA BEANS
 - AVOCADOS
 - BANANAS
 - PEANUTS
 - TURKEY
 - CHICKEN
 - EDAMAME

Raise Your Serotonin-Be Happy



Chocolate: High in magnesium and other 'feel good' nutrients



Leafy Greens: Energy enhancing



Walnuts: Loaded with Omega 3 and other brain nutrients



Epsom Salts: Calming



Water: Hydration increases energy and lowers stress



Cayenne Pepper Depression Reliever



Helps raise serotonin



Green Smoothie: For an energy zap



Smiling: Releases happy hormones



Walking: Clears the mind and raises Serotonin



Almonds: Brain food & magnesium



Oats: Eases Depression



Spend time in Nature

Melatonin

- Melatonin is a hormone that induces drowsiness
- Levels of melatonin begin climbing after dark
- Recommended for people with **insomnia** or **jet lag**, as well as for **shift workers**.
- High - Pistachios, Walnuts and almonds
- Medium – Cherries, Strawberries, Grapes and Wine



Dietary Sources and Bioactivities of Melatonin. Nutrients. 2017

Decoding Eating Patterns: Trends vs. Traditions

Keto Diet

- 70% to 75% of calories from fat, 20% from protein, and no more than 10% from carbohydrates.
- Cause ketosis.
- Can help lose 4/5 kgs or more in a short time
- The downside - potential rise in LDL (bad) cholesterol.
- Long-term use - linked to a higher risk of kidney stones, osteoporosis and gout



Carnivore diet

- Most ketogenic because its carb content is extremely low,
- Only eat meat, poultry, eggs, fish, and dairy.
- No vegetables, fruit, grains, legumes, seeds, and nuts.



-



Research Studies have linked DASH & Mediterranean diets to living longer by preventing chronic lifestyle diseases

DASH Eating Plan

The Benefits: Lowers blood pressure & LDL “bad” cholesterol.



Eat This



Limit This

	Vegetables		Fatty meats
	Fruits		
	Whole grains		Full-fat dairy
	Fat-free or low-fat dairy		
	Fish		Sugar sweetened beverages
	Poultry		
	Beans		Sweets
	Nuts & seeds		

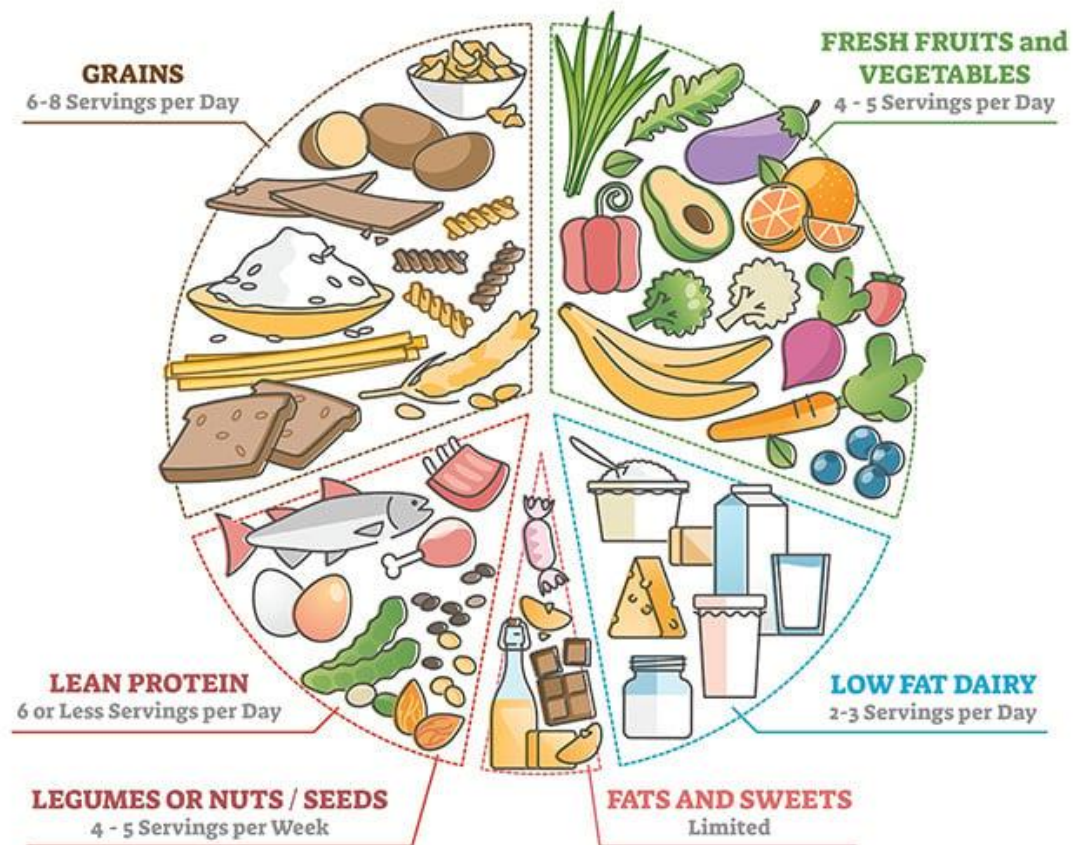
- Dietary Approaches to Stop Hypertension (DASH) diet

- Heavyweights

- Sustainable, Heart-healthy eating pattern suitable for long-term use

The DASH Diet

DIETARY APPROACHES TO STOP HYPERTENSION



Mediterranean Diet



- The 2024 JAMA study, tracked >25,000 women following Mediterranean diet for up to 25 years.
 - 23% lower risk of dying from any cause than those who did not
- Heart friendly Mediterranean diet:-
 - Plant based, olive oil, vegetables, fruits, whole grains, legumes, fish and nuts. Minimal to nil red meat.
 - High in antioxidants such as beta carotene, lycopene, and vitamins A, C, and E
- Lowers metabolic biomarkers – triglycerides, LDL
 - Lower insulin resistance – Reduces inflammation
- Better weight management

Fasting Diets – Intermittent Fasting

- Fasting—or purposely not eating food—dates back to ancient times, when people fasted in hopes of curing a disease or for religious reasons
- Alternate day fasting - significantly limiting calories for a full day
- 5:2 diet - you eat normally on five days, but restrict your calories on two nonconsecutive days



Intermittent Fasting

Eating during a limited time window (a form of fasting known as time-restricted eating) - only during the hours of 9 a.m. to 5 p.m.

Results from 23 *meta-analyses* of intermittent fasting studies - *eClinical Medicine* (2024)

- ***When you don't eat for extended hours, your body switches away from using carbohydrates for fuel and starts burning stored fat.***
- Included people - overweight or obese as well as some with diabetes or metabolic syndrome
- Control groups included people on calorie-restricted diets and habitual diets
- Researchers found good evidence that intermittent fasting can lower
 - waist circumference
 - triglycerides, LDL and total cholesterol
 - fasting insulin (the hormone that regulates blood sugar levels)
 - systolic blood pressure

Power Couples

- Beans and brown rice (Rajma Chawal) - complete protein source, containing all nine amino acids with good fibre
- Spinach and citrus - vitamin C needed to properly absorb iron.
- Tomatoes and olive oil - Healthy fats in olive oil or avocados improve uptake of lycopene in tomato, strong antioxidant and anticancer
- Turmeric and black pepper - Turmeric contains antioxidant curcumin, but poorly absorbed. Blending it with piperine, enhances its absorption, boosting anti-inflammatory effects
- There are many such examples

The Right Work Out



Aerobic Activity



150–300 mins moderate
or **75–150** mins vigorous
per week



Strength Training



2–3 times per week



Flexibility



2–3 times per week



Relaxation



Mix it up for maximum health benefits
Move more and sit less!

Exercise - Overcome Roadblocks and Get Moving

- Muscles contract and relax repeatedly - Stronger heart and lung function
- Improved health of blood vessels, better balance, and a healthier weight
- When muscles relax (Sitting), they take up very little sugar (glucose) from the blood, raising your risk of type 2 diabetes.
- Recent research - moderate intensity is also beneficial
- S (Specific), M (Measurable), A (Achievable), R (Realistic), T (Time) Goals
- Increases HDL



Aerobic Activities

- 150 - 300 minutes a week of moderate activity, or 75 - 150 minutes a week of vigorous movement, or an equivalent combination of both intensities spread throughout the week
- Moderate physical activity includes walking - Able to talk, not able to sing
- Vigorous exercise includes running, bicycling, and swimming - Finding talking difficult



Strength Training

- Isotonic – lifting a dumbbell from hip height to shoulder height, prompt muscles to shorten or lengthen to move the attached joint through its range of motion – ***good for heart.***
- Isometric - pushing against a wall or struggling to lift an extremely heavy weight - force muscles to work against fixed resistance, so no such shortening or lengthening occurs – ***builds muscle not the heart***



Flexibility exercises

- Stretching, yoga
- Reverse the shortening and weakening of muscles that typically occur as you age.
- Post-workout cool-down - Muscles become warm and pliable.



Relaxation Exercise

- Treat your mind and body to a mindful walk
- Mild yoga, and tai chi meld tension-melting movements with mental focus and meditation.



Interval Training

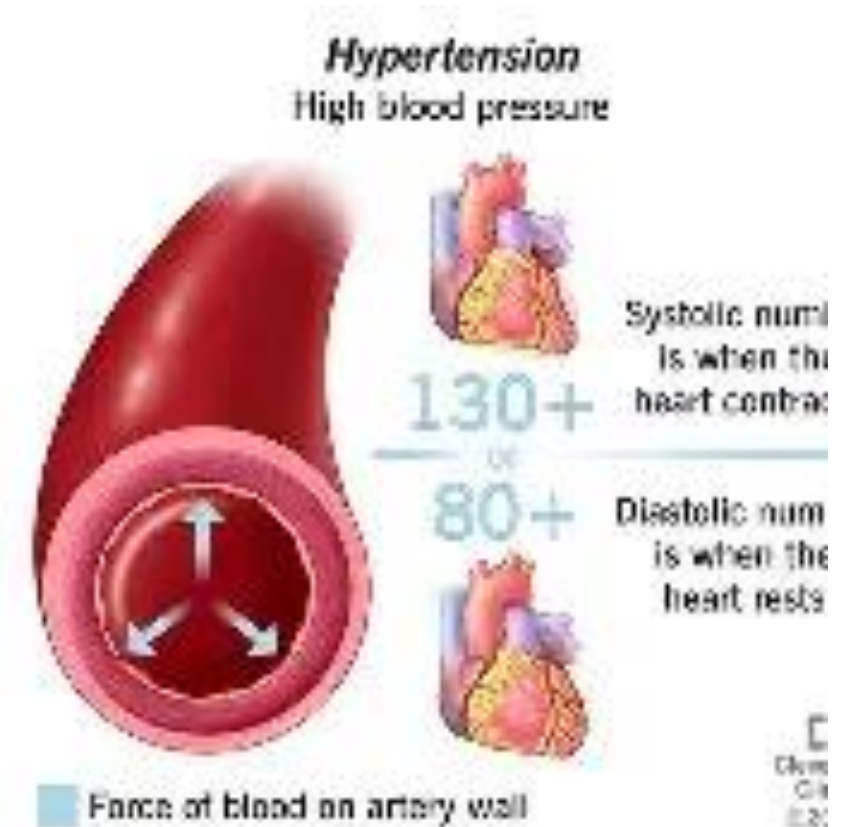
- Alternating short bursts of intense activity (vigorous) with periods of lighter recovery (moderate/rest). Alternating short bursts of high intense activity (vigorous) with periods of lighter recovery (moderate/rest).
- Creates Excess Post-Exercise Oxygen Consumption (EPOC) - continue burning calories for hours *after* the workout finishes.
- Improves VO2 Max (your body's ability to use oxygen) faster than steady jogging.

A close-up photograph of a hand holding a small, clear plastic bottle and pouring a clear liquid into the palm of another hand. The background is blurred, showing a person in a blue shirt. The image has a soft, warm color palette with a gradient from purple to orange.

Some Common Diseases and their Associated Dietary Risk Factors

Hypertension

- 2024 review in *Current Hypertension Reports* - white-coat hypertension is linked heart-related risks.
- Dietary Approaches to Stop Hypertension (DASH) diet with salt restriction & Exercise
- Exercise
- Reducing stress



Diabetes Primer

- Type 2 diabetes - the body is resistant to insulin
 - Glucose from the blood is not absorbed fully, causing blood sugar levels to rise.
 - Triggered by chronic inflammation
 - Belly fat – Waist Hip ratio

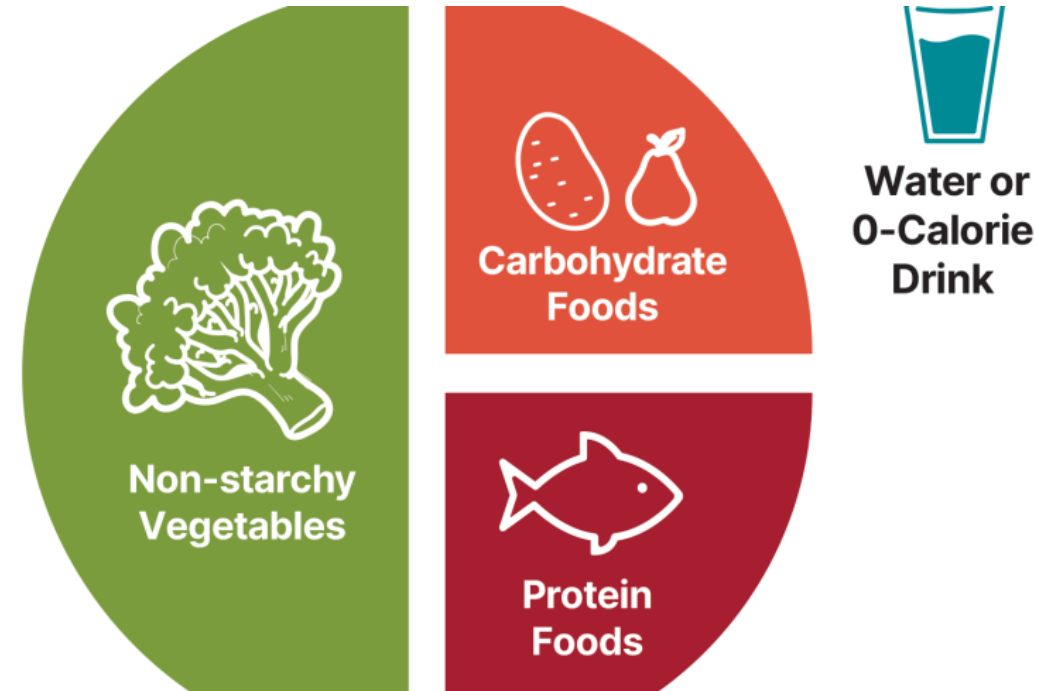


Diabetes Prevention Tips - 2026

Standards Of Care In Diabetes From

The American Diabetes Association

- Low GI and Low GL Food Groups, Mediterranean/DASH patterns
- Aerobic Exercise
- Strength training (such as lifting weights or using resistance bands) two to three times per week – improves insulin sensitivity
- Encourages drinking water instead of beverages sweetened with sugar or artificial sweeteners.

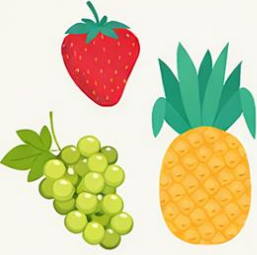
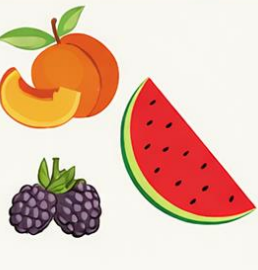


Obesity

- Cognitive Behavioral Therapy, Adequate Sleep, Stress Management
- 500–750 kcal/day deficit with Mediterranean/DASH patterns
- Aerobic + Strength Training
- **Pharmacotherapy:** (e.g., Semaglutide, Tirzepetide) for BMI > 30 or > 27.5 with comorbidities.



How to avoid IBS flare-ups with Food

staying on top of your Low FODMAP Diet					
Food	Veggies	Fruits	Protiens	Fats	Starches Cereals, Grains
Eat					
Avoid					

Prevent Dementia



Keeping systolic blood pressure close to 120 mm Hg may be best for brain health.



Apples, berries, grapes, and citrus fruits are among the flavonoid-rich foods that can reduce dementia risk.

Neurology in 2025



Daily olive oil consumption is linked to reduced risks of dying from dementia.

Supplements

- Fills Nutritional Gaps
- Addresses deficiencies like Iron deficiency, Vit D, B12 deficiencies
- Omega 3 from Fish Oils
- Side effects if consumed in high doses
- Herbal dietary supplements – associated with liver injuries in several studies

Table 1. Homeopathic or Over-the-Counter Drugs and Other Products Incorrectly Categorized as Dietary Supplements.^a

Product Category	Homeopathic Drugs	Over-the-Counter Drugs	Other Products That Are Not Supplements
Immunity or infection	Cough and cold remedies (Zarbee's, Hyland's)	—	—
Laxative	—	Calcium polycarbophil Castor oil Malt soup Methylcellulose Mineral oil	—
Pain or arthritis relief	Arnica Pokeweed	—	Clove oil Pain-relief gel (Sombra)
Sexual enhancement	—	Erectile dysfunction medication	—
Skin or hair health	Thuja occidentalis Acne treatments (Nature's Cure)	—	Tea tree oil Topical herbal products Various hair products (Nioxin, Nutrisol-Rf)
Sleep, sedation, or anxiolysis	Ignatia amara	—	—
Other specified products	Aconite Belladonna Calcarea carbonica Eupatorium or mistletoe Rhus toxicodendron Gripe water Pulsatilla intestinal formula Lomatium dissectum Homeopathic eyedrops Homeopathic sulfur eardrops Teething-relief remedies (Hyland's, Humphreys')	—	Cinnamon oil Peppermint oil Other oils (Olbas Oil)

Nutrition Facts Label

What's in it for you?

Nutrition Facts

8 servings per container

Serving size 2/3 cup (55g)

Amount per serving

Calories 230

% Daily Value*

Total Fat 8g 10%

Saturated Fat 1g 5%

Trans Fat 0g

Cholesterol 0mg 0%

Sodium 160mg 7%

Total Carbohydrate 37g 13%

Dietary Fiber 4g 14%

Total Sugars 12g

Includes 10g Added Sugars 20%

Protein 3g

Vitamin D 2mcg 10%

Calcium 260mg 20%

Iron 8mg 45%

Potassium 235mg 6%

* The % Daily Value (DV) tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.



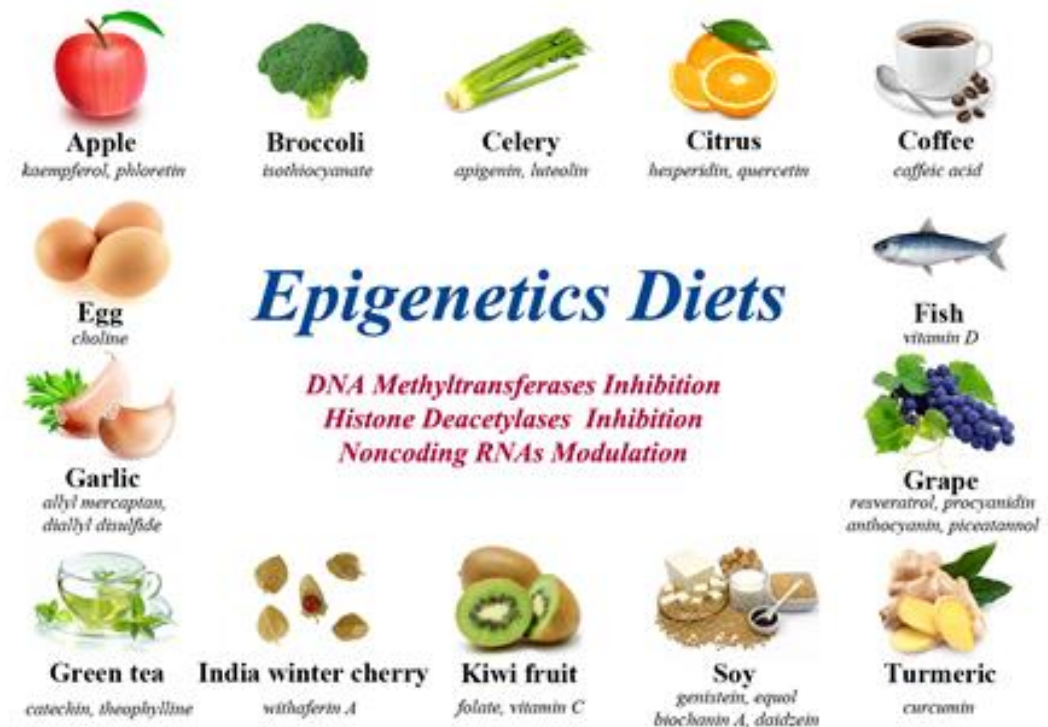
FAQs & Myths

Any
Questions?

I have a few

Do genes or lifestyle determine your health?

- Both your **genes** and **lifestyle** affect your health, with lifestyle and environment usually having a bigger impact.
- Scientists studying protective genes in long-lived people with unhealthy habits.
- **Epigenetics** explores chemical changes that turn genes on or off without changing DNA.
- These changes can be reversed, such as quitting smoking improving lung-related markers within a year.



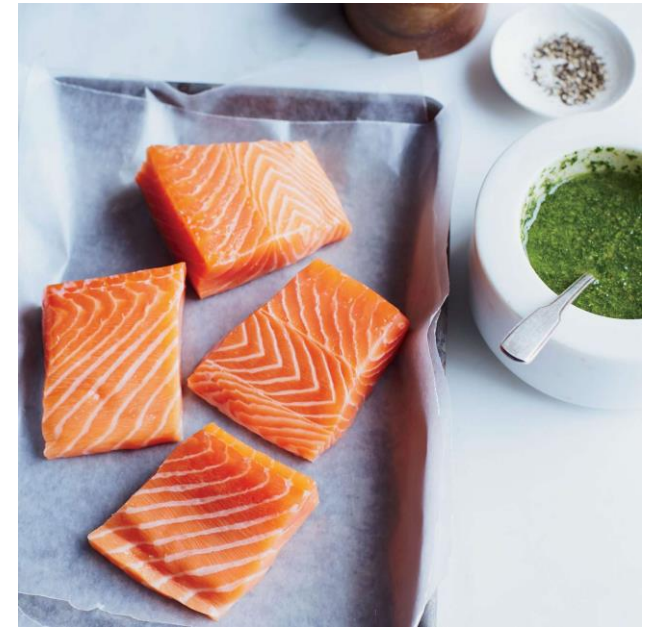
Is Sea Food a Good Choice?

American Heart Association - eat at least two servings of seafood per week

Use seafood to replace animal proteins - less saturated fat and more omega-3 fatty acids.

Farmed salmon and canned tuna are good alternatives to hamburgers and cold cuts

You can go even further down the aquatic food chain by eating seaweed



Seaweed is low in calories and high in heart-healthy phytonutrients

White Bread vs Brown Bread?

- White Bread - refined flour, Low in fiber, protein, vitamins, and minerals; high in carbohydrates (high glycemic index)
- Brown Bread –partly whole wheat and partly refined flour or maida, can be colored (caramelized)
- Whole Wheat Bread - Uses the entire grain, preserving germ and bran, Highest in protein, fiber, B vitamins, Vitamin E, magnesium, iron, phosphorus, zinc; lower glycemic index.

INGREDIENTS : Wheat Flour (Atta) (32%), Refined Wheat Flour (Maida) (30%), Water, Sugar, Wheat Protein (Gluten), Baker's Yeast, Iodized Salt, Malt, Soya Flour, Edible Vegetable Oil (Rice Bran and Soyabean), Class II Preservative (280, 281, & 282), Flour Treatment Agents (516), Emulsifier (472(e) & 481(i)), Acidity Regulator (260), Improver {1100 (i) & 1104} and Antioxidant (300).
(NUMBERS IN BRACKET AS PER INTERNATIONAL NUMBERING SYSTEM)
[CONTAINS SOYA AND GLUTEN]
ALL INGREDIENTS ARE OF 100% VEGETABLE ORIGIN.



Eat cheese, if you please

- *Slices of cheddar, crumbled feta, and shaved Parmesan - heartfriendly.*
- The fermentation produces vitamin K - help slow the buildup of harmful calcium deposits in the heart's arteries and valves.
- Blue cheese, feta cheese, cottage cheese and Parmesan are higher in sodium than other cheeses
- Processed "cheese" - contains added ingredients, such as preservatives and colorings.



Eating about 1.5 ounces of feta cheese per day is linked to a lower risk of cardiovascular disease

Are salt substitutes a smart choice?

Using a potassium-containing salt substitute instead of regular salt may lower blood pressure



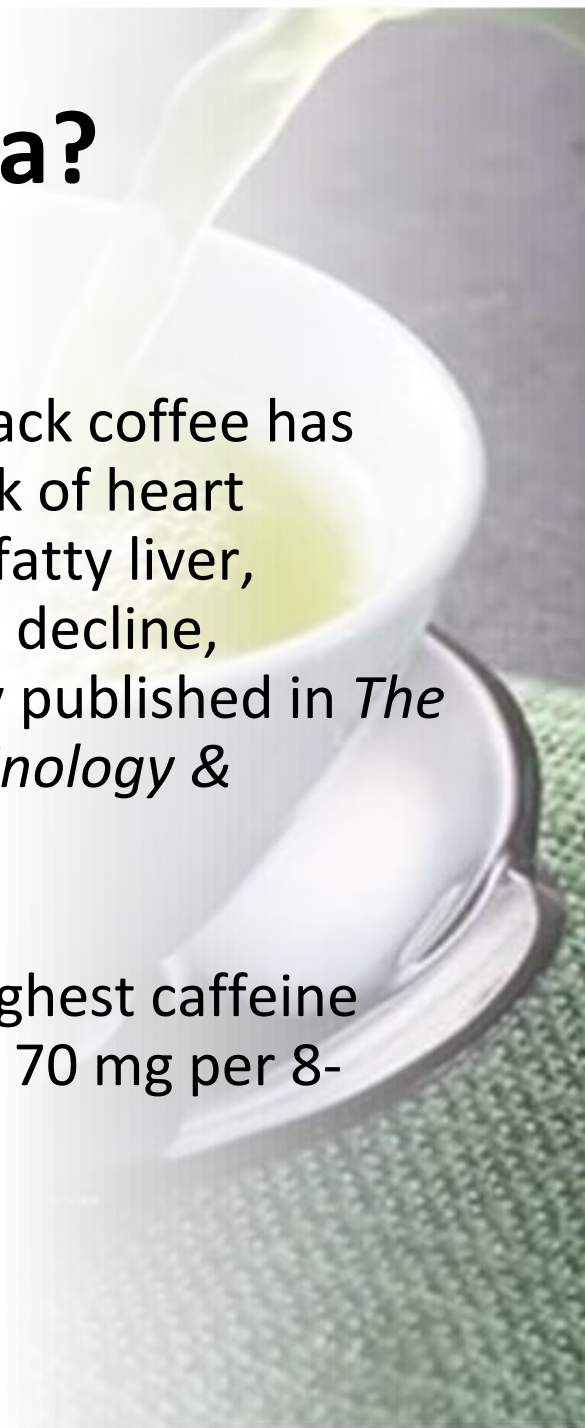
Cow's Milk – Healthy Choice

- Contains 60 to 70% SFA and 3 to 5% PUFA
- High quality proteins
- Calcium & Phosphorus
- Paneer contains almost same (70%) amount of SFA, Protein, Calcium
- Less (60%) in Curd from same quality milk
- Skimmed (1%) – recommended by AHA
- Double Toned (1.5%)
- Toned (3%)
- Ration – 4 to 5% fat



Coffee and Tea?

- Drinking unsweetened black coffee has been linked to a lower risk of heart disease, type 2 diabetes, fatty liver, depression, and cognitive decline, according to a 2024 study published in *The Journal of Clinical Endocrinology & Metabolism*.
- Green tea contains the highest caffeine levels, ranging from 40 to 70 mg per 8-ounce serving.



Are pollutants a growing concern?

- Air Pollution – linked to Heart and Lung Disease
- Chemical - Heavy metals (especially lead)
- Use 100% PFA (Poly Fluoro Alkyls) free cooking utensils
- Pesticides – in food chain – linked to cancer risks



Key Take Aways

The "Power Couples" (Food Synergy)

Functional Foods for Gut & Heart

The "Red Flag" List

"The 9 to 5 Window": Try Time-Restricted Eating

Mediterranean and DASH Diet



Thanks



The most powerful medicine
is at the end of your FORK,
not at the bottom of
your pill bottle.
FOOD is more powerful
than anything in your
medicine cabinet!

- Dr. Mark Hyman -